

STA 2023 – Final Exam Practice

Spring 2026

Question 1:

A public health researcher at UCF wants to study whether a new wellness program reduces students' stress levels.

The true average stress level of all UCF students is 65, but this value is unknown to the researcher. She selects 200 UCF students who volunteer to participate in the program. After 8 weeks, the average stress level of the sample decreases from 66 to 54.

(a) Identify the following:

- i. Population
- ii. Sample
- iii. Parameter
- iv. Statistic

(b) Classify each of the following variables as **qualitative** or **quantitative**. If quantitative, state whether it is **discrete** or **continuous**.

- i. Stress score (0–100 scale)
- ii. Student major (e.g., Nursing, Business, Engineering)
- iii. Number of hours students sleep per night
- iv. Whether a student participates in the program (Yes/No)

(c) Identify the level of measurement for each variable:

- i. Stress score
- ii. Student major
- iii. Number of hours of sleep

(d) Is this study an **observational study** or a **designed experiment**? Justify your answer.

(e) Can the researcher conclude that the wellness program **caused** the reduction in stress levels? Explain clearly.

- (f) Suppose instead the researcher randomly selected students from all UCF students using a student ID list.
- What type of sampling method is this?
 - Why is this method preferred over the original method used?

Question 2:

A professor records the exam scores, out of 100, for a group of students:

45, 52, 60, 63, 67, 70, 72, 75, 78, 85, 90, 95, 120

Instruction: Show all steps. Answers without supporting work will not receive full credit.

(a) Find the five-number summary:

- Minimum
- First quartile, Q_1
- Median, Q_2
- Third quartile, Q_3
- Maximum

(b) Construct a boxplot of the data. Clearly label:

- Q_1 , median, and Q_3
- Whiskers
- Any outliers

(c) Based on the boxplot, describe the shape of the distribution as symmetric, left-skewed, or right-skewed. Justify your answer.

Question 3:

A university survey shows the following information about 200 students:

- 120 students own a laptop
- 80 students own a tablet
- 50 students own both a laptop and a tablet

A student is selected at random.

Instruction: Show all steps. Answers without supporting work may not receive full credit.

- (a) Find the probability that the student:
- i. Owns a laptop
 - ii. Owns a tablet
 - iii. Owns both a laptop and a tablet
- (b) Find the probability that the student owns **at least one** of the two devices.
- (c) Find the probability that a student owns a **tablet given that the student owns a laptop**.
- (d) Are the events “owns a laptop” and “owns a tablet” **independent**? Justify your answer using probability calculations.
- (e) Suppose 3 students are selected at random **without replacement**. Find the probability that **all three students own a laptop**.

Question 4:

A game is played where a player rolls a fair die once.

- If the player rolls a 1 or 2, they lose \$2.
- If the player rolls a 3, 4, or 5, they win \$1.
- If the player rolls a 6, they win \$5.

Let X be the amount of money, in dollars, won in one play of the game.

Instruction: Show all steps. Answers without supporting work may not receive full credit.

(a) Construct the probability distribution of the random variable X .

(b) Find the expected value, $E(X)$.

(c) Find the standard deviation of X .

Now suppose the game is played 10 times independently. Let Y be the number of times the player wins \$1, that is, rolls a 3, 4, or 5.

(d) Find the probability that the player wins \$1 at least 3 times.

(e) Find the mean and standard deviation of Y .

Question 5:

The average weight of adult males in a population is

$$\mu = 180 \text{ pounds}$$

with a standard deviation of

$$\sigma = 20 \text{ pounds.}$$

A random sample of size

$$n = 64$$

adult males is selected.

Instruction: Show all steps. Answers without supporting work may not receive full credit.

- (a) Find the mean and standard deviation of the sampling distribution of the sample mean $\bar{X}$.
- (b) Find the probability that the sample mean weight is between 175 and 185 pounds.

Now consider a different scenario:

It is known that 30% of adults exercise regularly. A random sample of size

$$n = 100$$

adults is selected. Let $\hat{p}$ be the sample proportion of adults who exercise regularly.

- (c) Verify whether the normal approximation can be used for $\hat{p}$.
- (d) Find the mean and standard deviation of the sampling distribution of $\hat{p}$.
- (e) Find the probability that the sample proportion is greater than 0.35.

Question 6:

A university health office wants to estimate the proportion of students who support making flu vaccination clinics available on campus every semester. A random sample of **250 students** is selected, and **165 students** say they support the idea.

Instruction: Show all steps.

- (a) Construct a 95% confidence interval for the true proportion of all students who support having flu vaccination clinics on campus every semester.

Solution:

The formula for a confidence interval for one population proportion is:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

First, compute the sample proportion:

$$\hat{p} = \frac{x}{n} = \frac{165}{250} = 0.66$$

For a 95% confidence interval:

$$z_{\alpha/2} = 1.96$$

Substitute into the formula:

$$0.66 \pm 1.96 \sqrt{\frac{0.66(1 - 0.66)}{250}}$$

$$0.66 \pm 1.96 \sqrt{\frac{0.66(0.34)}{250}}$$

$$0.66 \pm 1.96 \sqrt{0.0008976}$$

$$0.66 \pm 1.96(0.02996)$$

$$0.66 \pm 0.0587$$

Therefore,

$$\text{Lower bound} = 0.66 - 0.0587 = 0.6013$$

$$\text{Upper bound} = 0.66 + 0.0587 = 0.7187$$

$$(0.6013, 0.7187)$$

Thus, the 95% confidence interval is approximately:

$$60.13\% \text{ to } 71.87\%$$

(b) Interpret the confidence interval in context.

Solution:

We are 95% confident that the true proportion of all students who support having flu vaccination clinics on campus every semester lies between **60.13%** and **71.87%**.